L1

unit: (1, 0) / $\begin{bmatrix}1&0\\0&1\end{bmatrix}$

zero: (0, 0) / $\begin{bmatrix}0&0\\0&0\end{bmatrix}$

im. unit: (0, −1) / $\begin{bmatrix}0&-1\\1&0\end{bmatrix}$

$(a,b) \longleftrightarrow \begin{bmatrix}a&-b\\b&a\end{bmatrix}$

$(a,b)\cdot(c,d)=(ac-bd,\ ad+bc)$

L2

modulus/ abs. value/radius: $\sqrt{x^2+y^2}$

$z\cdot\bar{z}=|z|^2=x^2+y^2$

$|z_1\cdot z_2|=|z_1|\cdot|z_2|$

$z_1\cdot z_2=r_1\cdot r_2\cdot(\cos(\theta_1+\theta_2)+i\cdot\sin(\theta_1+\theta_2))$

$z^n=r^n(\cos(n\theta)+i\cdot\sin(n\theta))$

$=(\cos\theta+i\cdot\sin\theta)^n$

find $\sqrt[n]{z}$: 化为 ←,

且 $Re(1)=1\cdot(\cos(\pi+2k\pi)+i\cdot\sin(\pi+2k\pi))$

L3

复数无法比较大小

3 pt.'s $\frac{d-b}{c-a}=\frac{f-d}{e-c}$ is colinear $\begin{pmatrix}(a,b)\\(c,d)\\(e,f)\end{pmatrix}$

3 complex pt.'s z_1, z_2, z_3 are colinear

$\Leftrightarrow \arg\left(\frac{z_1-z_2}{z_3-z_2}\right)=0$ or π $\Leftrightarrow imag(\sim)=0$

$\underbrace{([0,2\pi) \text{ or } (-\pi,\pi])}$ ↓

Principal argument $\arg(z)$

$\arg\left(\frac{z_1}{z_2}\right)=(\arg(z_1)-\arg(z_2)) \pmod{2\pi}$

↳ $\exists k\in\mathbb{Z}$, s.t. $\arg\left(\frac{\sim}{\sim}\right)-(\sim - \sim)=2k\pi$

LFT: linear fractional transformations $f(z)=\frac{az+b}{cz+d}$

If $c=0$: $\left(\frac{a}{d}\right)z+\left(\frac{b}{d}\right)$

If $c\neq0$: $\frac{az+b}{cz+d}=\frac{a}{c}-\frac{ad-bc}{c}\cdot\frac{1}{cz+d}$

L4

Extended complex plane: $\mathbb{C}\cup\{\infty\}=\overline{\mathbb{C}}$

~~complex~~ function $f(z): \overline{\mathbb{C}}\to\overline{\mathbb{C}}$

LFT

$$= \begin{cases}\frac{az+b}{cz+d} & \text{if } \sim \\ \infty & \text{if } z=-\frac{d}{c} \\ \frac{a}{c} & \text{if } z=\infty\end{cases}$$

$\frac{c}{0}=\infty,\ \frac{c}{\infty}=0$

LFT is a bijection on $\overline{\mathbb{C}}$

Thm.: Inverse LFT $f(z)=\frac{1}{z}$ is a LFT

→ preserves circles or 直线

map 圆/直线 to 圆/直线

Thm.: $\exists$unique LFT,

s.t. $f(z_1)=w_1$, $\sim=w_2$, $\sim=w_3$

L5

$\exp(z)=e^z=\sum_{k=0}^{\infty}\frac{z^k}{k!}$

$e^z=e^x(\cos y+i\cdot\sin y)$

$|z-w|=\sqrt{(x-x')^2+(y-y')^2}$

$|z_1+z_2|\le|z_1|+|z_2|$

Open Disc 以z_0为中心: $D(z_0,r)=\{z\in\mathbb{C}: |z-z_0|<r\}$

以∞为中心: $\{z\in\mathbb{C}: |z|>R\}$

开放 open 集 S: $\forall z_0\in S$, $\exists r>0$, s.t. $D(z_0,r)\subseteq S$

A seq. conv. → ∞: $\forall R>0$, $\exists N\in\mathbb{N}$, s.t. $\forall k\ge N$, $|z_k|>R$

A seq. conv. → $L\in\mathbb{C}$

$\Leftrightarrow$ Re(seq.) 和 Im(seq.)

$\xrightarrow{conv.}$ Re(L) $\xrightarrow{conv.}$ Im(L)

A seq. is Cauchy $\Leftrightarrow \forall\varepsilon>0, \exists N\in\mathbb{N}$, s.t. $\forall m,n\ge N$, $|z_m-z_n|\le\varepsilon$

$\Leftrightarrow$ seq. is conv.

series: $\sum_{k=0}^{\infty}z_k=\lim_{n\to\infty}\left(\sum_{k=0}^{n}z_k\right)\xrightarrow{exist}\{z_k\}_\infty$ conv.

$\{z_k\}_\infty$ conv. $\Rightarrow \lim_{n\to\infty}z_n=0$

$\Leftarrow$ abs. conv.

ratio test $\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right|=L$

$L>1\to\{a_k\}_\infty$ div.

$L<1\to\sim$ conv.

$L=1\to$ nothing

$\sum_{l=0}^{k}\left(\frac{z_1^{l}}{l!}\cdot\frac{z_2^{k-l}}{(k-l)!}\right)=\frac{(z_1+z_2)^k}{k!}$

$e^{i\theta}=\cos\theta+i\sin\theta$

L6

$\sin(z)=z-\frac{z^3}{3!}+\frac{z^5}{5!}-\cdots$

$\cos(z)=1-\frac{z^2}{2!}+\frac{z^4}{4!}-\cdots$

$\underline{\sinh(z)}=z+\frac{z^3}{3!}+\frac{z^5}{5!}+\cdots=\frac{e^y-e^{-y}}{2}$

$\cosh' y=\sinh y$

$\sinh' y=\cosh y$

$\underline{\cosh(z)}=1+\frac{z^2}{2!}+\frac{z^4}{4!}+\frac{z^6}{6!}+\cdots=\frac{e^y+e^{-y}}{2}$

$e^{iz}=1+iz+\frac{i^2z^2}{2!}+\cdots$

$\underline{i}=\cos\frac{\pi}{2}+i\cdot\sin\frac{\pi}{2}=e^{i\cdot\frac{\pi}{2}}$

$\cos^2 z+\sin^2 z=1$

☆ $e^z=w \Leftrightarrow z=\underline{Log(w)}$

$Log(w)=\ln|w|+i\cdot(\arg(w)+2\pi k)$

若为principal log 则去掉

计算角度常用 arctan$\left(\frac{虚}{实}\right)$

计算 $\underline{a^z = e^{\ln|a| \cdot (x+iy)}} = \cos(\ln|a| \cdot x) + i \cdot \sin(\ln|a| \cdot y)$

$\cos(n\theta) = \frac{e^{in\theta} + e^{-in\theta}}{2}$

$$\frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y}\right)$$

L12

Winding number: of a closed path: $\underbrace{\frac{\text{total change in 角度}}{2\pi}}_{\in \mathbb{Z}}$ (含方向!!)

change in 角度 (angle) $= [\underbrace{f_1(z_1) - f_1(z_0)}_{\text{change in argument}}] + [f_2(z_2) - f_2(z_1)]$

z_1 → 中, z_0 → 始, z_2 → 尾, z_1 → 中

Inverse func: $g(x,y) = \overline{g(x,y)}$

$g'(x_0,y_0)$ is invetible $\Rightarrow \exists$ open set U, V, s.t. $g: U \to V$ is bij. and inverse of g is diff.

g is holomorphic $\Rightarrow g'(z_0)$ is rep. by $\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \Rightarrow \begin{bmatrix} a & -b \\ b & a \end{bmatrix}^{-1} = \frac{1}{a^2+b^2}\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$

power: $\underline{z^w = e^{w \cdot \log(z)}}$

$e^{i \cdot \log(i)} = i^i = e^{-\frac{\pi}{2}}$

L11

~~复三角函数~~

$F(x,y) \xrightarrow{\text{output}}$ set → 其一: $f(z)$ ($\in F(z)$)

e.g. $df = (x+y)dz + (x-y)d\bar{z}$

$f(x,y) = x+iy$

$\Rightarrow \frac{\partial f}{\partial z} = x+y$, $\frac{\partial f}{\partial \bar{z}} = x-y$

path/curve:

cont. $\gamma(t): [a,b] \to \mathbb{C}$

Summation formula:

$$z^{b} + z^{a+b} + \dots + z^{na+b} = \frac{z^b[1 - z^{(n+1)a}]}{1 - z^a}$$

L7

$f(z) = u(x,y) + i \cdot v(x,y)$ is cont.

$\Leftrightarrow$ u 和 v 都 cont.

$f(z)$ is diff. $\Leftrightarrow \lim\limits_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$ 存在

$\Rightarrow$ ~~let $p = (x_0, y_0)$,~~

$$\begin{cases} u_x(x_0, y_0) = v_y(\sim) \\ u_y(\sim) = -v_x(\sim) \end{cases}$$

L8

Holomorphic

$f(z)$ is ~ $\Leftrightarrow$ $\exists$ neighbor N, s.t. $\forall z \in N$, f is c. diff at z

$f(z)$ is complex diff. at $z_0 = x_0 + iy_0$

$$\Leftarrow \begin{cases} (u(x,y), v(x,y)) \text{ is diff. at } (x_0, y_0) \\ u_x(x_0, y_0) = v_y(\sim) \\ u_y(\sim) = -v_x(\sim) \end{cases}$$

u_x, u_y, v_x, v_y are $\begin{cases} \text{well-def. at } (\sim) \\ \text{cont. in a neighbor of } (\sim) \end{cases}$

$f(z)$ is entire $\Leftrightarrow$ f 在 $\mathbb{C}$ 上所有 z 中 c. diff.

L9

$f(z)$ is conformal at $z_0 \in \mathbb{C}$

$\Leftarrow f$ is diff. at z_0 & $f'(z_0) \neq 0$

$\Rightarrow \arg\left(\frac{\frac{d}{dt} f(\gamma_1(t_1))}{\frac{d}{dt} f(\gamma_2(t_2))}\right) = \sim = \arg\left(\frac{\gamma_1'(t_1)}{\gamma_2'(t_2)}\right) = \theta$

(角的大小、方向均保!)

Regions R_1 和 R_2 are conformally equiv.

$\Leftrightarrow \exists$ bij. $\underbrace{\text{map}}_{\text{holomorphic}}$ $f(z): R_1 \to R_2$

Harmonic func. $\Rightarrow f(z)$ is 2次 diff. & 偏导均 cont.

$f(z)$ is harmonic $\Leftrightarrow \begin{cases} u_{xx} + u_{yy} = 0 \\ v_{xx} + v_{yy} = 0 \end{cases}$ Laplace equation

Real-v. $u(x,y)$ is har. $\Leftrightarrow \underline{u_{xx} + u_{yy}} = 0$

$= \Delta u$

$u(x,y) + i \cdot v(x,y)$ is holomorphic

$\Rightarrow v(x,y)$ is a harmonic conjugate of $u(x,y)$

L10

$\underline{v}_p = (p, \underline{v})$ ($\underline{v}$ that emanates from p)

Tangent space $T_p = \{\underline{v}_p : \underline{v} \in \mathbb{R}^2\}$

$(p, \underline{v}_1) + (p, \underline{v}_2) = (p, \underline{v}_1 + \underline{v}_2)$

$a \cdot (p, \underline{v}) = (p, a\underline{v})$

$p \neq q \Rightarrow T_p \neq T_q$

Vector field: a func. that selects $\underline{v}_p$ for every p

T_p^*: the dual space of T_p (func: $T_p \to \mathbb{R}$)

$\forall$element $\in T_p^*$ is linear functional/covector

Directional deri.: $(r\underset{(\to x)}{\underline{e}_1} + s\underset{(\to y)}{\underline{e}_2})_p$ 的: $(\nabla f)_p \cdot (r\underline{e}_1 + s\underline{e}_2)$

$\phi_f(\underline{v}_p) = (\nabla f)_p \cdot \underline{v}_p \overset{?}{=} (df)_p$

is a l. functional at p

differential form: a func. that selects a l. func. for $\forall$point(x,y)

e.g. df

complex-valued diff. form: $\alpha dx + \beta dy$ ($\alpha, \beta \in \mathbb{C}$)

$(df = du + idv)$

$(dz)_p = (dx)_p + i(dy)_p$,

$(d\bar{z})_p = (\sim)_p - i(\sim)_p$

$(dx)_p = \frac{dz + d\bar{z}}{2}$, $(dy)_p = \frac{dz - d\bar{z}}{2i}$

$df = \frac{\partial f}{\partial \bar{z}}(dz + d\bar{z})$

$\underline{\frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right) = 0} \Leftrightarrow \begin{cases} u_x = v_y \\ u_y = -v_x \end{cases}$